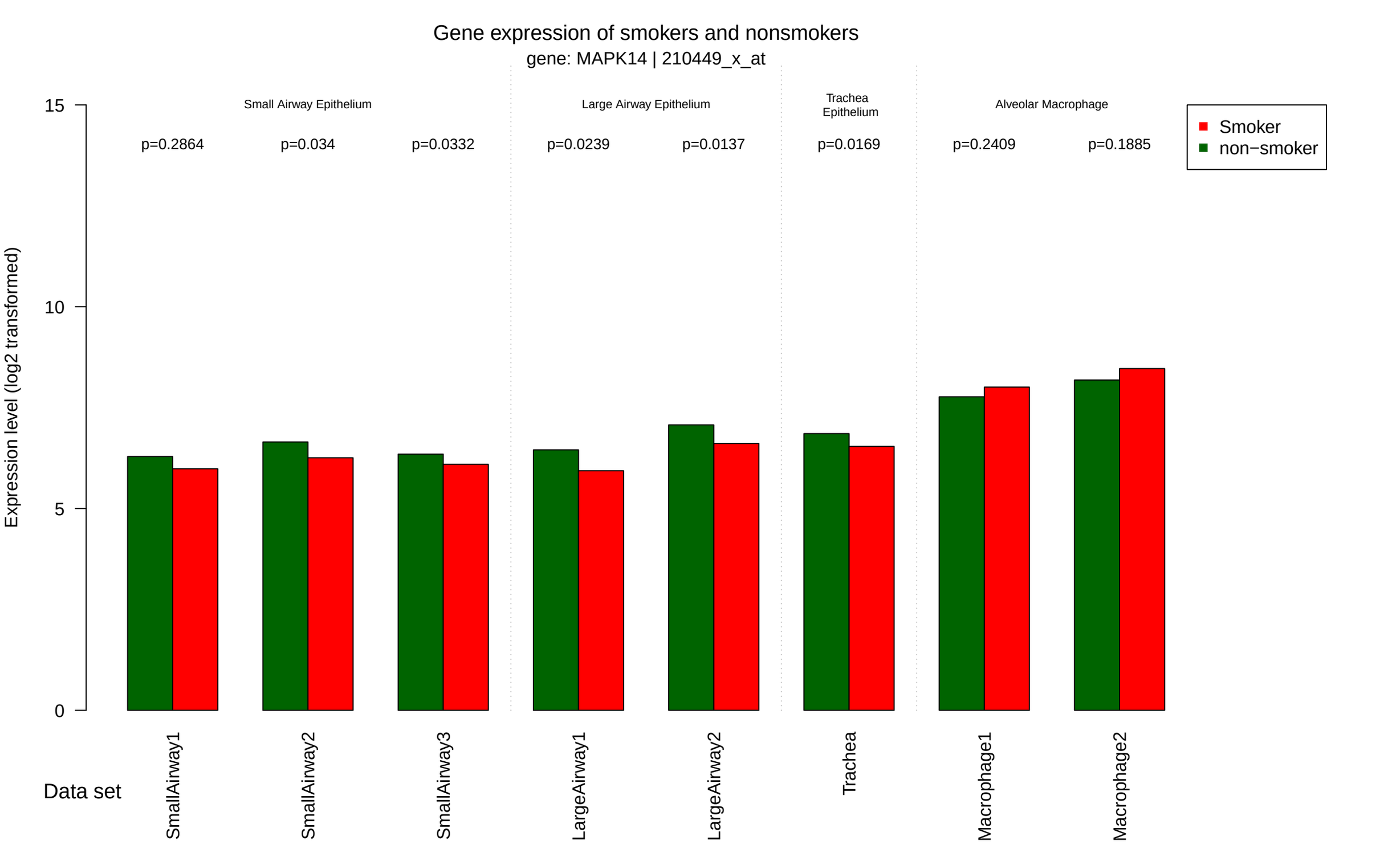
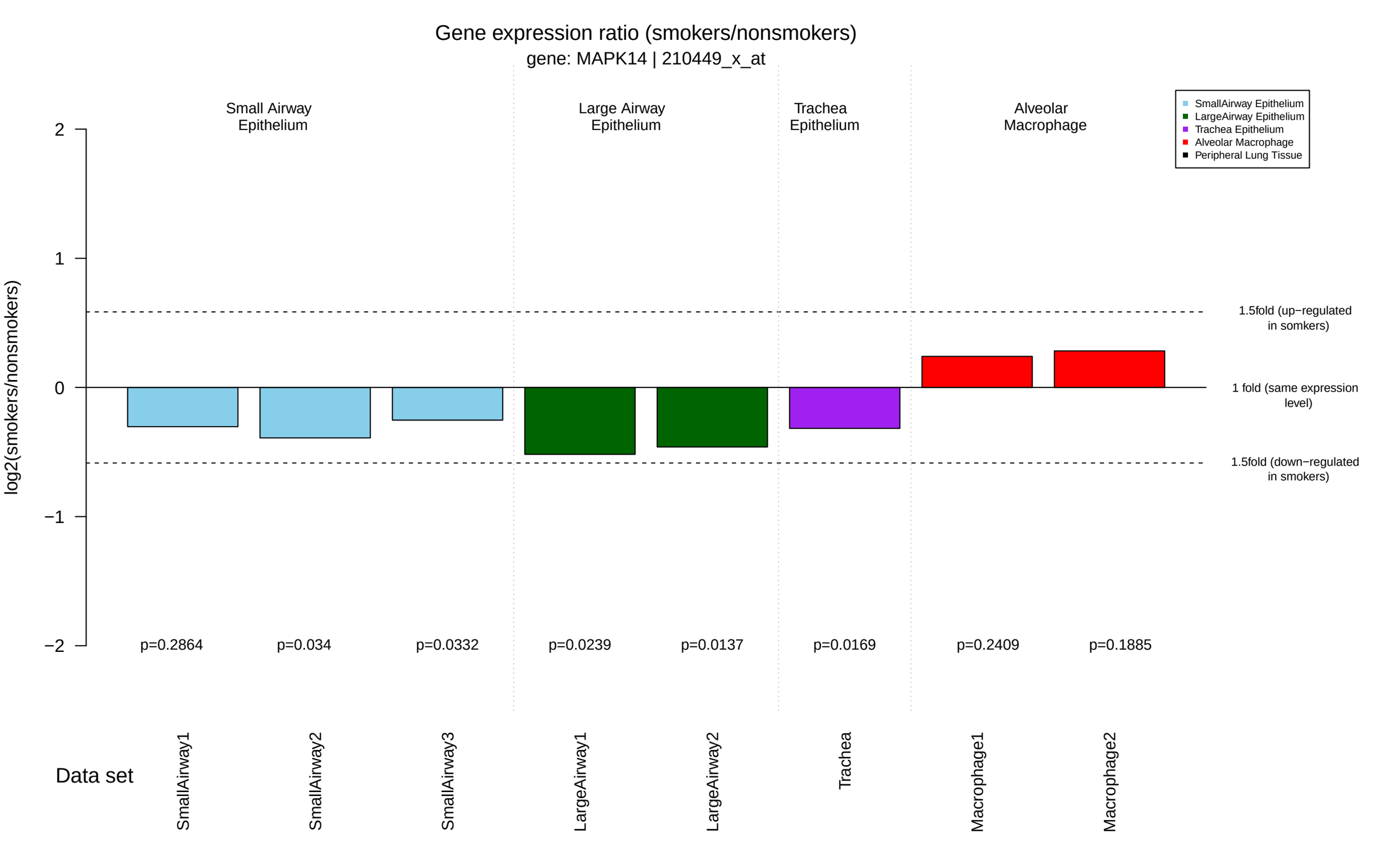
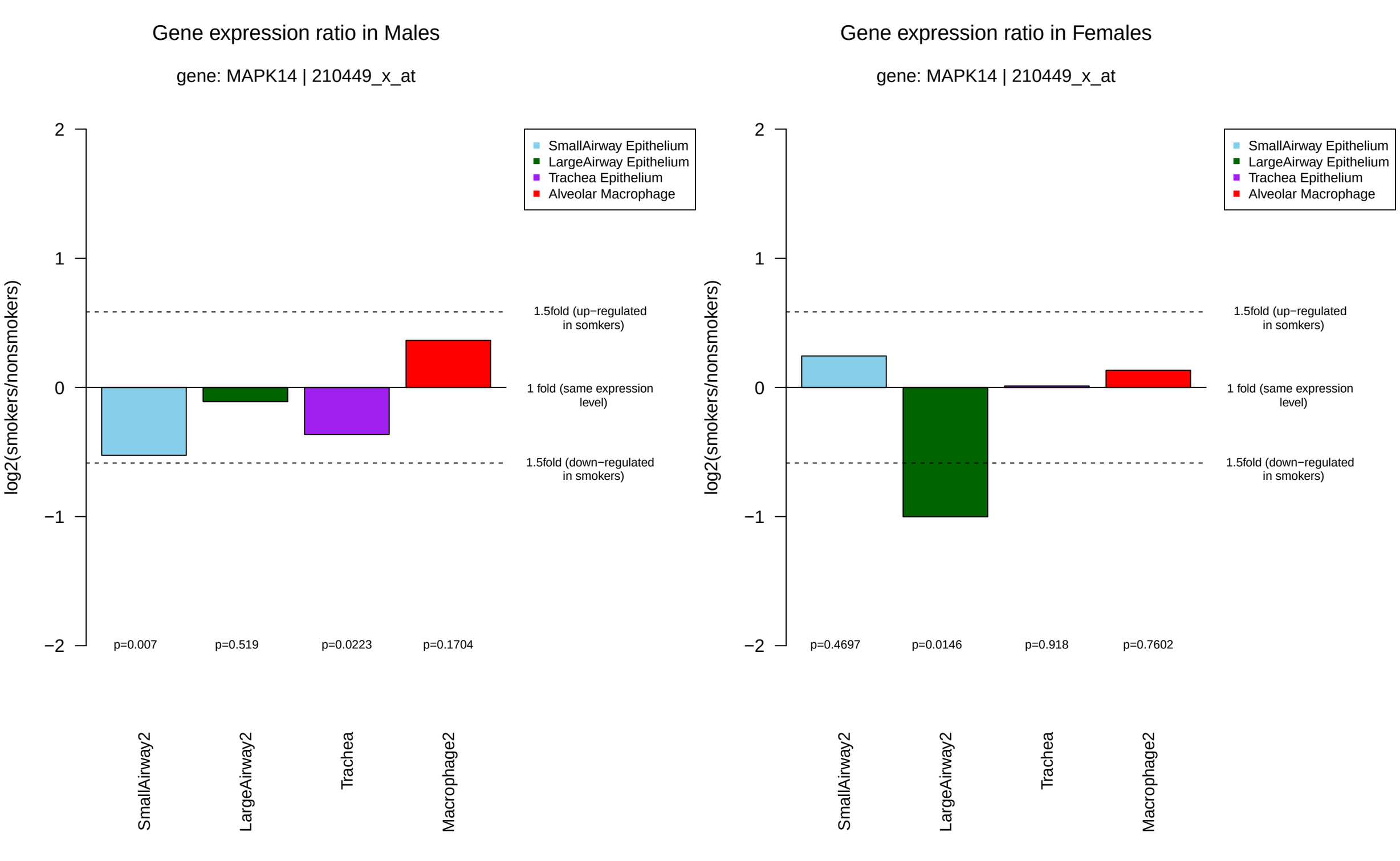
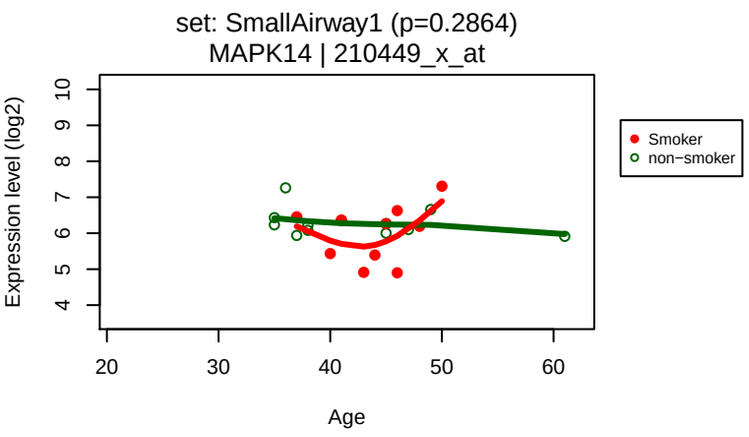
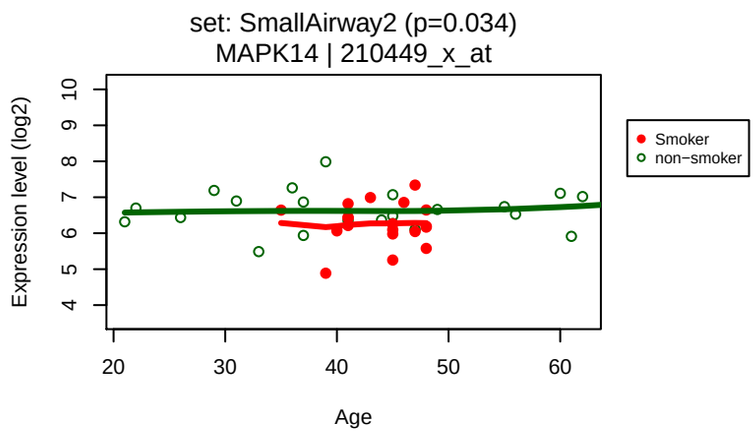
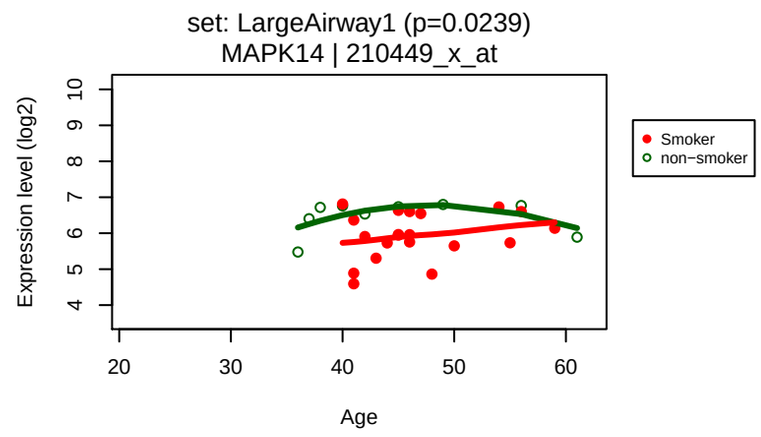

Gene expression of smokers and nonsmokers
gene: MAPK14 | 210449_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2864
p=0.034
p=0.0332
p=0.0239
p=0.0137
p=0.0169
p=0.2409
p=0.1885
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

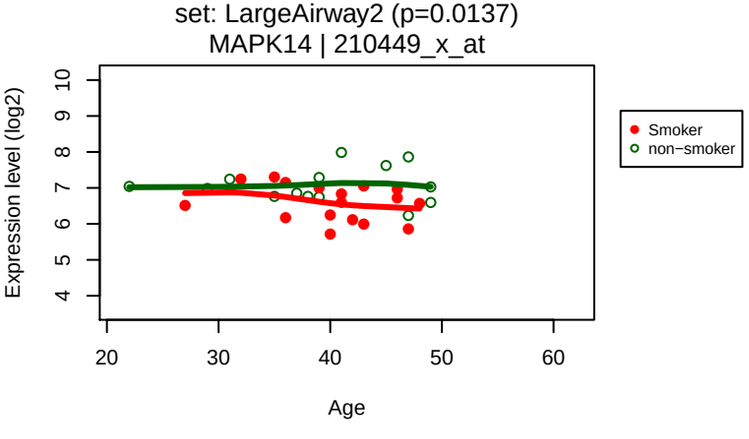

Gene expression ratio (smokers/nonsmokers)
gene: MAPK14 | 210449_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2864
p=0.034
p=0.0332
p=0.0239
p=0.0137
p=0.0169
p=0.2409
p=0.1885
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

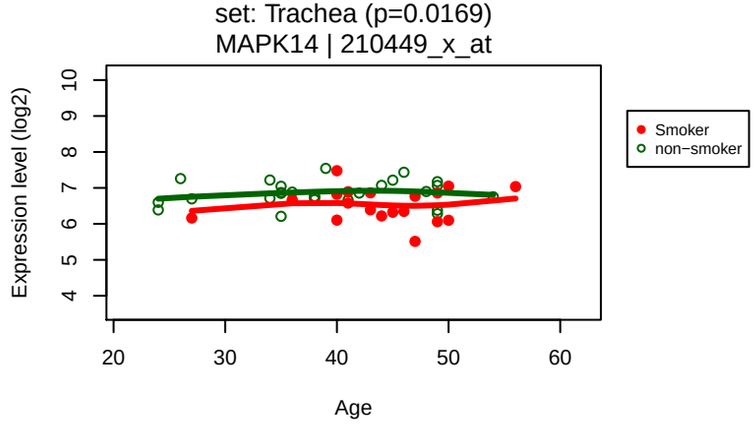

Gene expression ratio in Males
gene: MAPK14 | 210449_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.007
p=0.519
p=0.0223
p=0.1704
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MAPK14 | 210449_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4697
p=0.0146
p=0.918
p=0.7602
SmallAirway2
LargeAirway2
Trachea
Macrophage2

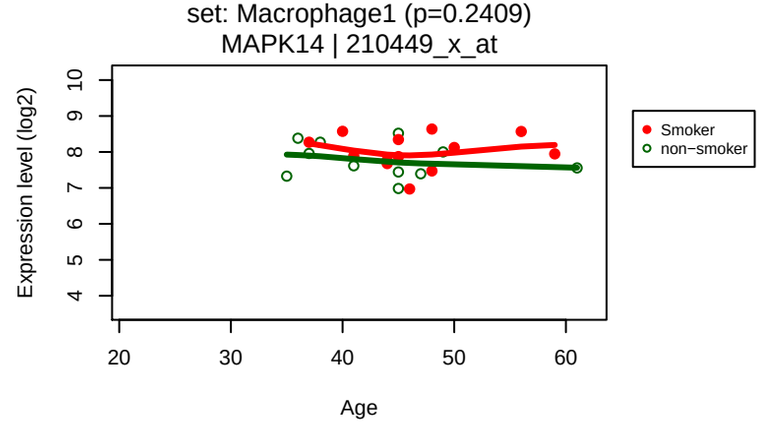

set: SmallAirway1 (p=0.2864)
MAPK14 | 210449_x_at
Expression level (log2)
Age
Smoker
non−smoker

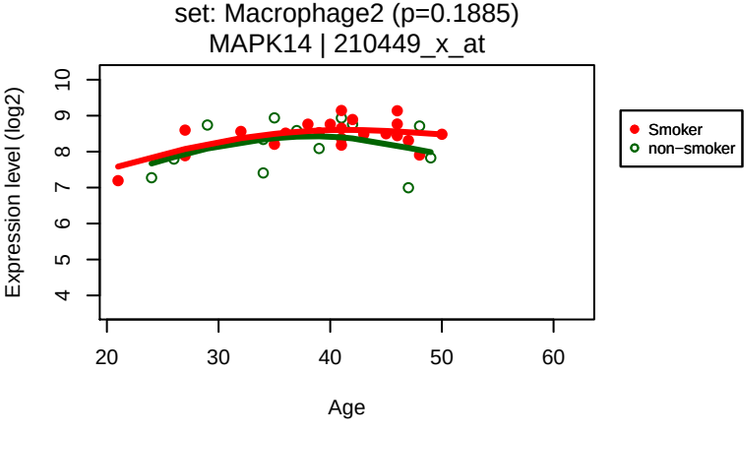

set: SmallAirway2 (p=0.034)
MAPK14 | 210449_x_at
Expression level (log2)
Age
Smoker
non−smoker

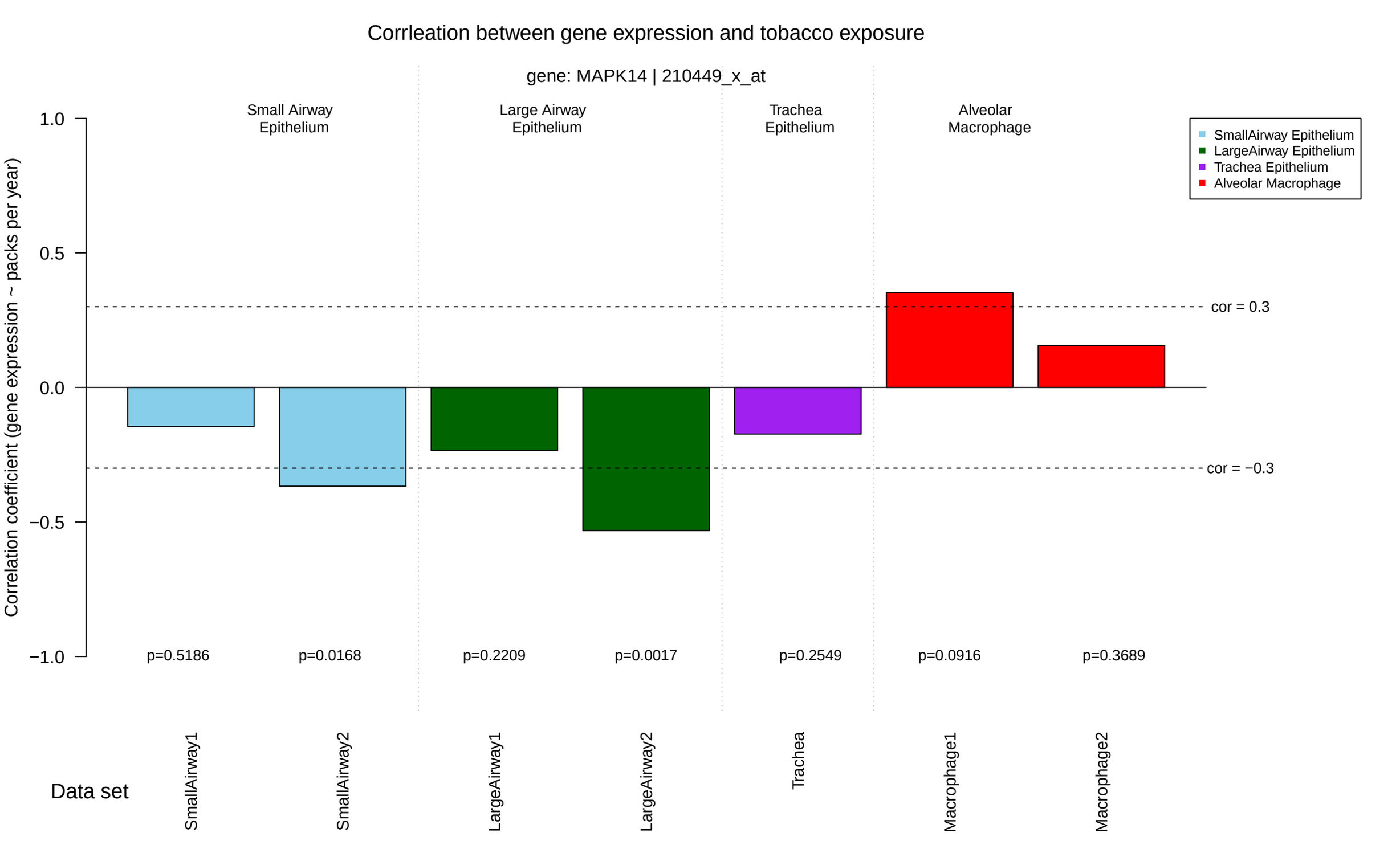

set: LargeAirway1 (p=0.0239)
MAPK14 | 210449_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0137)
MAPK14 | 210449_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0169)
MAPK14 | 210449_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2409)
MAPK14 | 210449_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1885)
MAPK14 | 210449_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MAPK14 | 210449_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5186
p=0.0168
p=0.2209
p=0.0017
p=0.2549
p=0.0916
p=0.3689
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2